

Hockey Game State Analysis using Markov Chains

Cameron Fietz
ECE 463 – A01

Problem Statement

The objective of this project is to predict the likelihood of an NHL team winning the game based on the current game state by implementing Markov chains. The current game state would consist of a combination of different factors (i.e. goal differential, shot differential, time remaining etc.), each of which would influence the win probability. The results would then be simulated using a Markov process, considering only the current state and outputting the probability of winning the game.

A similar topic was explored in the paper “Predictive Modeling and Analysis of Hockey Using Markov Chains” [1]. The authors created two state-based stochastic models that utilized Markov processes. The first model incorporated shot and goal differential relative to the opposing team to approximate the probability that the home team would win depending on the state they are currently in at a given time in the game. A second model built upon the first and explored the effect different man-advantage situations had on the win probability.

Methods

This project will take input from publicly available datasets and filter the data into a set of critical features. For example, MoneyPuck [2] and NHL.com [3] provide advanced analytics for the various situations (5on5, 5on4, 5on3, etc), including useful metrics such as:

- goals for/against
- shots for/against
- etc.
- expected goals for/against
- shot attempts for/against

To derive the probabilities for each metric, the stats of the winning team for each game from previous seasons in the dataset would be analyzed. This would result in a histogram that would be converted into a cumulative probability distribution for each metric.

A Markov process will then be used to simulate the probability that the team will win the game provided a game state. The current game state would include a weighted combination of these metrics. An example game state would follow a pattern similar to:

(man-advantage situation, time remaining, goal differential, expected goals differential, shot differential, shot attempt differential)

Thus, for a game that is tied with 30 minutes left and is played at 5on5, the state corresponds to:

$(0, 30, 0, x_4, x_5, x_6, x_7)$, where values x_4, x_5, x_6, x_7 correspond to each differential, respectively.

The effectiveness of this process could then be tested by applying it to real single-game situations and comparing the predictions with the true result.

References

- [1] A. Harsy *et al.*, “Predictive Modeling and Analysis of Hockey Using Markov Chains,” *Mathematics and Sports*, 2022. [Online]. Available: <https://janeway.uncpress.org/ms/article/id/1219>.
- [2] “Data,” *MoneyPuck*, 2024. [Online]. Available: <https://moneypuck.com/data.htm>.
- [3] “Stats,” *NHL.com*, 2024. [Online]. Available: <https://www.nhl.com/stats>.